Baire category degenerates for pointwise-convergence metrics on non-autonomous systems, and how equicontinuity restores it

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Abstract

Balibrea and Rucká [arXiv:2410.10288] asked whether distributional chaos of type 1 (DC1) is a generic property of convergent non-autonomous systems on the interval, and emphasized the metrics of their class ϱ_1 — such as the integral metric — which, unlike the supremum metric, admit genuinely pointwise-but-not-uniformly convergent systems. The uniform case was settled affirmatively in the companion paper [2]. Here we settle the ϱ_1 case, in an unexpected way: for every metric of integral type, the space of continuous surjections of the interval, and with it every associated space of pointwise convergent non-autonomous systems, is *meager in itself*. Such a space is not Baire, and every subset of it is simultaneously meager and comeager, so Baire genericity carries no information: the question itself degenerates. The proof rests on a modulus-of-continuity decomposition together with a spike perturbation that is small in measure, hence small in any integral-type metric, yet uniformly separated from all maps of bounded modulus. We then show that equicontinuity is exactly the missing ingredient: on each equicontinuity stratum the integral and supremum metrics are uniformly equivalent and the stratum of systems is complete — hence Baire — so genericity questions become well-posed again; there, the genericity of DC1 reduces to the density of horseshoe maps within the stratum of single maps, which fails for nonexpansive strata (these contain no DC1 pair whatsoever) and which we formulate as a precise open problem in general, identifying two quantitative obstructions. Together with [2], this closes the genericity problem of Balibrea and Rucká over their full range of metrics.

1 Introduction

Let $I = [0,1]$ and let $C_s(I)$ denote the continuous surjections $I \to I$. A non-autonomous system is a sequence $f_{1,\infty} = (f_n)_{n\geq 1} \subseteq C_s(I)$ acting by the compositions $f^n_{1,\infty} = f_n \circ \cdots \circ f_1$; distributional chaos of type 1 (DC1) is recalled in Section 2. In [1], Balibrea and Rucká studied the genericity of DC1 among *convergent* systems. They formulated the problem for an arbitrary metric ρ on $C_s(I)$, with the system space carrying $\rho_{\mathcal{F}}(f_{1,\infty}, g_{1,\infty}) = \sup_n \rho(f_n, g_n)$, and they divided the metrics into two classes: the class ϱ_2 (e.g. the supremum metric $\rho_{\sup}$), for which every convergent system converges uniformly, and the class ϱ_1 (e.g. the integral metric $\rho_{\mathrm{int}}(f,g) = \int_0^1 |f-g|\,dx$), for which convergence of maps is compatible with genuinely pointwise, non-uniform convergence of the system. They proved that DC1 systems are dense for every metric, that DC1 is generically *absent* on the Cantor set, that on the interval the DC1 systems form a set that is neither open nor closed, and they stated the residuality question as an open problem.

In the companion paper [2] we proved that for the supremum metric DC1 is generic: the DC1 systems contain the open dense set of systems whose limit map carries a strict horseshoe. That settles the ϱ_2 side. The present paper settles the ϱ_1 side, where the named difficulty — pointwise versus uniform convergence — actually lives, and where, as [1] already observed, the ambient spaces fail to be complete. Our first main result shows the failure is as severe as possible.

Theorem 1.1 (Degeneracy)**.** *Let ρ be a metric of integral type on $C(I,I)$ (Definition 2.1; the class contains ρ_{int} and all L^p metrics, $1 \le p < \infty$, and is a formalization of the class ϱ_1 of [1]). Then:*

(i) $(C(I,I),\rho)$ and $(C_s(I),\rho)$ are meager in themselves;

(ii) every space $\mathcal{F}$ of non-autonomous systems $(f_n) \subseteq C_s(I)$ that is closed under modification of the first coordinate within $C_s(I)$ — in particular the space of pointwise convergent systems with continuous limit, with metric $\sup_n \rho(f_n, g_n)$ — is meager in itself;

(iii) consequently these spaces are not Baire, and in each of them every *subset is simultaneously meager and comeager.*

In particular, both statements "DC1 is generic" and "the absence of DC1 is generic" hold vacuously in the ϱ_1 setting: Baire category does not distinguish them. The genericity problem of [1] therefore has exactly one substantive reading, the uniform one, which is resolved in [2]. This phenomenon seems worth recording in its own right: the pointwise-convergence metrics do not merely lose completeness, they lose the entire Baire framework.

The second part of the paper identifies what survives. The maps responsible for meagerness are spikes of unbounded steepness; freezing the steepness removes them.

Theorem 1.2 (Restoration on equicontinuity strata)**.** *Let μ be a modulus of continuity and let $C_s^\mu := \{f \in C_s(I) : \ |f(x) - f(y)| \le \mu(|x-y|)\ \forall x, y\}$, and let $\mathcal{F}^\mu$ be the space of systems $(f_n) \subseteq C_s^\mu$ that converge pointwise, with the metric $\sup_n \rho(f_n, g_n)$ for any integral-type ρ. Then:*

(i) on C_s^μ the metrics ρ and $\rho_{\sup}$ are uniformly equivalent; the space $(C_s^\mu, \rho_{\sup})$ is compact, every pointwise convergent system in it converges uniformly, and $\mathcal{F}^\mu$ is a complete, hence Baire, metric space, with the same topology for every integral-type ρ satisfying (U);

(ii) if the maps with a strict horseshoe are $\rho_{\sup}$-dense in C_s^μ, then DC1 is generic in $\mathcal{F}^\mu$: the DC1 systems contain a relatively open dense subset;

(iii) if $\mu(t) \le t$ (nonexpansive stratum), then no *system in $\mathcal{F}^\mu$ has even a single DC1-scrambled pair — in particular the horseshoe class is empty and DC1 is* not *generic there.*

Thus the Baire framework, which Theorem 1.1 destroys globally, is fully restored on equicontinuity strata, and on each stratum the genericity of DC1 reduces to a horseshoe-density question *about single maps* of the stratum. By (iii) the answer is negative for nonexpansive strata; we identify in Section 4 two quantitative obstructions (modulus saturation and the surjectivity-repair budget) that make the density question delicate for a general modulus, and we state it as a precise open problem (Problem 4.2).

2 Preliminaries

For $f_{1,\infty} = (f_n) \subseteq C_s(I)$, $x, y \in I$, the lower and upper distributional functions ψ_{xy}, ψ^*_{xy} are defined as in [1, 2] through the orbit distances $d_j = |f^j_{1,\infty}(x) - f^j_{1,\infty}(y)|$: $\psi_{xy}(t) = \liminf_n \frac{1}{n}\#\{j < n : d_j < t\}$ and ψ^* with lim sup. A pair is DC1-scrambled if $\psi^* \equiv 1$ and $\psi(t_0) = 0$ for some $t_0 > 0$; a system is DC1 if some uncountable set is pairwise DC1-scrambled.

Lebesgue measure on I is λ. A *modulus of continuity* is a nondecreasing $\mu : [0,1] \to [0,\infty)$ with $\mu(0^+) = 0$; ω_f denotes the optimal modulus of f.

Definition 2.1 (Integral-type metrics)**.** *A metric ρ on $C(I,I)$ is of* integral type *if:*

(V) (vanishing on small sets) *There is a function φ with $\varphi(0^+) = 0$ such that for all $f, g \in C(I,I)$,*

$$\rho(f,g) \;\le\; \varphi\big(\lambda(\{x : f(x) \neq g(x)\})\big).$$

(P) (positivity on spread-out differences) *For all $s, \ell \in (0,1]$ there is $c(s,\ell) > 0$ such that whenever $f, g \in C(I,I)$ satisfy $|f(x) - g(x)| \ge s$ for all x in some interval of length ℓ, then $\rho(f,g) \ge c(s,\ell)$.*

We will also refer to the following property, used only in Section 4:

(U) (*domination by the uniform metric*) $\rho(f,g) \le \Phi\big(\rho_{\sup}(f,g)\big)$ for some Φ with $\Phi(0^+) = 0$.

Property (V) is the defining property of the class ϱ_1 in [1, Remark 1] (metrics for which restrictions to small-measure sets carry vanishing weight, so that no single point carries positive distance); property (P) is a minimal nondegeneracy requirement, without which ρ could not separate maps that differ on a fixed interval. For $\rho_p(f,g) = \big(\int |f-g|^p\big)^{1/p}$, $1 \le p < \infty$, all three hold with $\varphi(\delta) = \delta^{1/p}$, $c(s,\ell) = s\,\ell^{1/p}$ and $\Phi(s) = s$. The supremum metric satisfies (P) and (U) but not (V). Theorem 1.1 uses only (V) and (P); Theorem 1.2 assumes (V), (P) and (U).

Lemma 2.2 (Spikes)**.** *Let $f \in C_s(I)$, let $M \in \mathbb{N}$, and let $\delta > 0$. There exist $g \in C_s(I)$ and an interval J_g of length $\ell_g > 0$ such that:*

(a) $\lambda(\{f \neq g\}) < \delta$*;*

(b) for every $h \in C(I,I)$ with $\omega_h(1/M) \le \frac{1}{100}$ there is an interval $J_h \subseteq I$ of length ℓ_g with $|g - h| \ge \frac{1}{32}$ pointwise *on J_h.*

Proof. Pick an interval $A = [a, a+r]$ with $r < \min\big(\delta, \frac{1}{2M}\big)$ and small enough that $\omega_f(r) \le \frac{1}{100}$. Write $c := f(a)$ and assume $c \le \frac{1}{2}$; the case $c > \frac{1}{2}$ is symmetric (reflect the construction in the line $y = \frac{1}{2}$). Put

$$d := \max\Big(0,\, c - \tfrac{1}{50}\Big), \qquad v := c + \tfrac{1}{3} \;\le\; \tfrac{5}{6}.$$

Divide A into eight consecutive closed intervals $A_1, \dots, A_8$ of length $r/8$, and define $g := f$ off A, while on A the map g is piecewise linear with the following values: on A_1 it interpolates from $f(a)$ to d; on A_2 it is constant d (*low plateau*); on A_3 it interpolates from d to v; on $A_4 \cup A_5$ it is constant v (*high plateau*); on A_6 it interpolates from v to d; on A_7 it is constant d; on A_8 it interpolates from d to $f(a+r)$. All these values lie in $[0,1]$, so $g \in C(I,I)$, and $\lambda(\{f \neq g\}) \le r < \delta$, giving (a).

Surjectivity. Off A we have $g = f$, so $g(I) \supseteq f(I \setminus A)$, and the only values possibly lost are those of $f(A)$. Since $\omega_f(r) \le \frac{1}{100}$, $f(A) \subseteq [c - \frac{1}{100}, c + \frac{1}{100}] \subseteq [d, v]$ (lower end: if $d = c - \frac{1}{50}$

then $c - \frac{1}{100} \geq d$; if $d = 0$ then $f(A) \subseteq [0,1]$ has lower values $\geq 0 = d$). But $g(A_3) = [d, v]$, so $g(A) \supseteq f(A)$ and $g(I) = I$: $g \in C_s(I)$.

The gap (b). Set $J_g := A_4$ and $\ell_g := r/8$. Let $h \in C(I, I)$ with $\omega_h(1/M) \leq \frac{1}{100}$. Every point of A_2 and every point of $A_4 \cup A_5$ lie within distance $r \leq \frac{1}{2M} < \frac{1}{M}$ of each other, so h varies by at most $\frac{1}{100}$ between A_2 and A_4. Suppose, for contradiction, that $|h - g| < \frac{1}{32}$ at some point $x \in A_2$ *and* at some point $y \in A_4$. Then

$$|h(x) - h(y)| \;\geq\; |g(y) - g(x)| - \tfrac{2}{32} \;=\; (v - d) - \tfrac{1}{16} \;\geq\; \tfrac{1}{3} - \tfrac{1}{50} - \tfrac{1}{16} \;>\; \tfrac{1}{4} \;>\; \tfrac{1}{100},$$

contradicting $|x - y| \leq \frac{1}{M}$. Hence $|g - h| \geq \frac{1}{32}$ pointwise on all of A_2 or on all of A_4; in either case (b) holds with $J_h \in \{A_2, A_4\}$ of length $\ell_g = r/8$. □

3 Degeneracy: proof of Theorem 1.1

Proof of Theorem 1.1. Let ρ satisfy (V) and (P). For $M \in \mathbb{N}$ set

$$D_M := \Big\{ f \in C(I, I) : \; \omega_f(1/M) \leq \tfrac{1}{100} \Big\}, \qquad C(I, I) = \bigcup_{M \geq 1} D_M,$$

the union holding because every continuous map on a compact interval is uniformly continuous. We claim each closure $\overline{D_M}$ (in ρ) has empty interior; this proves (i) for $C(I, I)$, and intersecting with $C_s(I)$ proves it for $C_s(I)$ since the spike construction preserves surjectivity.

Let $f \in C(I, I)$ (resp. $C_s(I)$) and $\varepsilon > 0$. By (V) choose $\delta > 0$ with $\varphi(\delta) < \varepsilon$ and let g be the spike modification of Lemma 2.2; then $\rho(f, g) \leq \varphi(\delta) < \varepsilon$. By Lemma 2.2(b) and property (P), every $h \in D_M$ satisfies $\rho(g, h) \geq c\big(\frac{1}{32}, \ell_g\big) > 0$. Hence $g \notin \overline{D_M}$, and since ε was arbitrary, $\overline{D_M}$ contains no ball: it is nowhere dense. Thus $C(I, I)$ (and $C_s(I)$) is a countable union of nowhere dense sets.

For (ii), let $\mathcal{F}$ be a space of systems closed under modification of the first coordinate, with metric $\rho_{\mathcal{F}} = \sup_n \rho(f_n, g_n)$. Write $\mathcal{F} = \bigcup_M \{f_{1,\infty} \in \mathcal{F} : \; f_1 \in D_M\}$. Each piece is contained in $\pi_1^{-1}(\overline{D_M})$ where $\pi_1(f_{1,\infty}) = f_1$ is 1-Lipschitz, so its closure is contained in the closed set $\pi_1^{-1}(\overline{D_M})$; and the latter has empty interior in $\mathcal{F}$: given $f_{1,\infty} \in \mathcal{F}$ and $\varepsilon > 0$, replace f_1 by its spike modification $g_1 \in C_s(I)$ (the system $g_{1,\infty} = (g_1, f_2, f_3, \dots)$ stays in $\mathcal{F}$ by hypothesis — modifying one map does not affect pointwise convergence or the limit), so that $\rho_{\mathcal{F}}(f_{1,\infty}, g_{1,\infty}) = \rho(f_1, g_1) < \varepsilon$ while $\rho_{\mathcal{F}}\big(g_{1,\infty}, \pi_1^{-1}(D_M)\big) \geq \rho(g_1, D_M) \geq c(\frac{1}{32}, \ell_{g_1}) > 0$. Thus $\mathcal{F}$ is meager in itself.

For (iii): if a metric space X is meager in itself, it is not Baire by definition; moreover for every $A \subseteq X$, A is meager (a subset of the meager set X) and $X \setminus A$ is meager, i.e. A is comeager. □

Corollary 3.1. *For every integral-type metric, in the space of pointwise convergent non-autonomous systems on I with continuous limit, the set of DC1 systems and its complement are both comeager. Baire genericity is uninformative in the ϱ_1 setting, and the genericity problem of [1] is substantive precisely in its uniform reading, resolved in [2].*

Remark 3.2. *Theorem 1.1 explains, structurally, the examples of [1, Section 3]: the systems with steeper and steeper tents that converge pointwise to the identity while carrying positive entropy are exactly the unbounded-modulus phenomena on which the decomposition $\bigcup_M D_M$ has no grip. It also shows that no choice of ϱ_1-metric can repair this: properties (V) and (P) alone force meagerness-in-itself. Any meaningful theory of typical pointwise-convergent systems must therefore either change the notion of typicality (e.g. prevalence) or restrict the space, as in the next section.*

4 Restoration on equicontinuity strata: proof of Theorem 1.2

Throughout this section ρ is an integral-type metric satisfying also (U), and μ is a modulus with $C_s^\mu \neq \emptyset$ (equivalently $\mu(1) \geq 1$, since a surjection of I has oscillation 1).

Proof of Theorem 1.2(i). C_s^μ is uniformly bounded and equicontinuous, hence totally bounded in $\rho_{\sup}$ by Arzelà–Ascoli, and it is $\rho_{\sup}$-closed in $C(I)$: a uniform limit of maps with modulus μ has modulus μ, and surjectivity passes to uniform limits of self-maps of a compact space. Hence $(C_s^\mu, \rho_{\sup})$ is compact.

By (U), the identity $(C_s^\mu, \rho_{\sup}) \to (C_s^\mu, \rho)$ is continuous; it is a bijection from a compact space onto a metric (hence Hausdorff) space, so it is a homeomorphism, and the two metrics, being topologically equivalent metrics on a compact space, are *uniformly* equivalent: there are nondecreasing Φ_1, Φ_2 with $\Phi_i(0^+) = 0$ and

$$\rho \leq \Phi_1(\rho_{\sup}), \qquad \rho_{\sup} \leq \Phi_2(\rho) \qquad \text{on } C_s^\mu \times C_s^\mu. \tag{1}$$

Taking suprema over n in (1) (the Φ_i are monotone) shows that the system metrics $\sup_n \rho(f_n, g_n)$ and $\sup_n \rho_{\sup}(f_n, g_n)$ are uniformly equivalent on $(C_s^\mu)^{\mathbb{N}}$: they have the same topology, the same Cauchy sequences, and the same Baire property. We may therefore work with $\rho_{\sup}$ throughout.

Within the equicontinuous family C_s^μ, pointwise convergence implies uniform convergence: given ε, pick a finite δ-net $x_1, \dots, x_m$ with $\mu(\delta) < \varepsilon/3$; then $|f_n(x) - f(x)| \leq 2\mu(\delta) + \max_i |f_n(x_i) - f(x_i)|$. So $\mathcal{F}^\mu$ is the set of $\rho_{\sup}$-convergent sequences in $(C_s^\mu)^{\mathbb{N}}$. The space $\big((C_s^\mu)^{\mathbb{N}}, \sup_n \rho_{\sup}\big)$ is complete (uniformly bounded sup-metric over complete factors), and the subset of convergent sequences is closed in it: if $f_{1,\infty}^{(k)} \to g_{1,\infty}$ in $\sup_n \rho_{\sup}$ with each $(f_n^{(k)})_n$ convergent, then for $\varepsilon > 0$ choose k with $\sup_n \rho_{\sup}(f_n^{(k)}, g_n) < \varepsilon/3$ and N with $\rho_{\sup}(f_n^{(k)}, f_m^{(k)}) < \varepsilon/3$ for $n, m \geq N$; then $\rho_{\sup}(g_n, g_m) < \varepsilon$, so (g_n) is convergent. Hence $\mathcal{F}^\mu$ is complete, and Baire. □

We first record the horseshoe class and the conditional genericity statement, then discuss why the hypothesis of (ii) is delicate.

Definition 4.1. *$\mathcal{O} \subseteq C_s(I)$ is the set of maps carrying a* strict two-branch horseshoe*: disjoint nonempty closed intervals $A_1 < A_2$ and $\theta > 0$ with $g(A_i) \supseteq [\min A_1 - \theta, \max A_2 + \theta] \subseteq (0, 1)$ for $i = 1, 2$. As in [2], $\mathcal{O}$ is $\rho_{\sup}$-open (an interval image containing an interval with margin θ persists with margin $\theta/2$ under $\rho_{\sup}$-perturbations below $\theta/2$), every $g \in \mathcal{O}$ has topological entropy at least $\log 2$, and every map in $\mathcal{O}$ is DC1 by Schweizer–Smítal [3].*

Proof of Theorem 1.2(ii). By (i) we may work with $\rho_{\sup}$ and its system metric, and $\mathcal{F}^\mu$ is a complete metric space. The limit-map function $\lambda(f_{1,\infty}) = \lim_n f_n$ is 1-Lipschitz from $\mathcal{F}^\mu$ to $(C_s^\mu, \rho_{\sup})$, so $U := \lambda^{-1}(\mathcal{O} \cap C_s^\mu) \cap \mathcal{F}^\mu$ is relatively open. It is dense: given $(f_n) \in \mathcal{F}^\mu$ with limit f and $\varepsilon > 0$, the hypothesis of (ii) provides $h \in \mathcal{O} \cap C_s^\mu$ with $\rho_{\sup}(h, f) < \varepsilon/2$; pick N with $\rho_{\sup}(f_n, f) < \varepsilon/2$ for $n > N$ and pass to the system $(f_1, \dots, f_N, h, h, \dots) \in \mathcal{F}^\mu$, which is ε-close and has limit h. Every system in U is uniformly convergent (by (i)) to a limit map carrying a strict horseshoe, hence of entropy at least $\log 2$, hence DC1 by [3]; by Štefánková's inheritance theorem [4] the system itself is DC1. Thus the DC1 systems contain the relatively open dense set U and are residual in $\mathcal{F}^\mu$. □

The horseshoe-density hypothesis: two obstructions and an open problem

For the supremum stratum — no modulus constraint — the density of $\mathcal{O}$ in $C_s(I)$ was proved in [2] by implanting a steep zigzag near a fixed point. Inside a stratum C_s^μ the same strategy meets two quantitative obstructions, which we record because they delimit any future proof.

Saturation. The implanted map v differs from the base map u by a perturbation ζ supported near the implantation site. For a pair (x, y) with x in the site and y far from it, $|v(x) - v(y)| \le |u(x) - u(y)| + \|\zeta\|_\infty$, and if u *saturates* its modulus along such pairs ($|u(x) - u(y)| = \mu(|x - y|)$, which mollification does not destroy), every nonconstant perturbation overruns the budget. Any density proof must therefore first create slack at all scales — for instance by the rescaling $u \mapsto (1 - \gamma)u + \gamma/2$ — and such rescalings destroy surjectivity.

The repair budget. Restoring surjectivity after a rescaling by $1-\gamma$ requires moving some value back to 0 (and to 1). Within C^μ a correction of height $\gamma/2$ built from the profile $c\,\mu(|x - x_0|)$ needs $c\,\mu(\text{radius}) \ge \gamma/2$; the modulus budget available for corrections is of order $\gamma\mu$, so the correction radius must satisfy $\mu(\text{radius}) \gtrsim \frac{1}{2}$, i.e. the radius is of order $\mu^{-1}(\frac{1}{2})$ — *not* small. Large-support corrections interact with the implanted horseshoe and with each other, and for moduli with $\mu(1)$ close to 1 they may not fit in $[0, 1]$ at all. The two requirements — slack everywhere, surjectivity at the endpoints — thus compete for the same budget.

These obstructions are quantitative, not absolute: they disappear, for example, on strata where every map admits a point of strict sub-saturation at the relevant scales. We do not know the answer in general and consider the following the main question left open by this paper.

Problem 4.2. *For which moduli μ is the strict-horseshoe class $\mathcal{O} \cap C_s^\mu$ dense in $(C_s^\mu, \rho_{\sup})$ — equivalently (by Theorem 1.2(ii)), for which strata is DC1 generic in $\mathcal{F}^\mu$? By Theorem 1.2(iii) the answer is negative for $\mu(t) \le t$. Is it positive for every non-Lipschitz concave modulus? For Lipschitz strata $\mu(t) = Lt$, is it positive for every $L > 1$, with horseshoes produced at the level of an iterate (where the Lipschitz budget compounds to L^n), or supplied by a crookedness dichotomy (maps saturating the budget at every scale should already carry horseshoes)?*

Proof of Theorem 1.2(iii). Let $\mu(t) \le t$, so every f_n is nonexpansive: $|f_n(x) - f_n(y)| \le |x - y|$. Then for any x, y the orbit distances $d_j = |f_{1,\infty}^j(x) - f_{1,\infty}^j(y)|$ are nonincreasing in j, hence converge to some $\ell \ge 0$. If (x, y) were DC1-scrambled, then $\psi_{xy}^*(t) = 1$ for all $t > 0$ forces $\liminf d_j = \ell < t$ for every $t > 0$, i.e. $\ell = 0$; but then $d_j \to 0$, so for every $t_0 > 0$ one has $d_j < t_0$ eventually, whence $\psi_{xy}(t_0) = 1 \ne 0$. No DC1 pair exists. □

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Declaration on generative AI

During the preparation of this work the author used generative AI tools to accelerate drafting and revision. All mathematical claims, proofs, and bibliographic information were subsequently reviewed and edited by the author, who takes full responsibility for the content of the manuscript.

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